

SCIENCE ALL SETS 2023

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CHAPTER: 1 - Chemical Reactions and Equations (Chemistry)

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YEAR: 2023

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Study the following chemical reaction: $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}\uparrow$. The reducing agent in this reaction is:

OPTIONS: (A) Na (B) H₂O (C) NaOH (D) H₂

ANSWER: (A) Na

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an example of an endothermic process?

OPTIONS: (A) Formation of slaked lime (B) Decomposition of vegetable matter into compost (C) Dissolution of ammonium chloride in water (D) Digestion of food in our body

ANSWER: (C) Dissolution of ammonium chloride in water

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: To balance the following chemical equation, the values of the coefficients x and y respectively are: $x\text{Pb(NO}_3)_2 \rightarrow 2\text{PbO} + y\text{NO}_2 + \text{O}_2$

OPTIONS: (A) 2, 4 (B) 2, 2 (C) 2, 3 (D) 4, 2

ANSWER: (A) 2, 4

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When zinc reacts with sodium hydroxide, the product formed is:

OPTIONS: (A) Sodium oxide (B) Sodium zincate (C) Zinc hydroxide (D) Zinc oxide

ANSWER: (B) Sodium zincate

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In order to balance the following chemical equation, the values of x, y and z respectively are: $\text{MnO}_2 + x\text{HCl} \rightarrow \text{MnCl}_2 + y\text{H}_2\text{O} + z\text{Cl}_2$

OPTIONS: (A) 6, 2, 2 (B) 4, 1, 2 (C) 4, 2, 1 (D) 2, 2, 1

ANSWER: (C) 4, 2, 1

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A metal ribbon X burns in oxygen with a dazzling white flame forming a white ash Y. The correct description of X, Y and the type of reaction is:

OPTIONS: (A) X = Ca, Y = CaO, Type of reaction = Decomposition (B) X = Mg, Y = MgO, Type of reaction = Combination (C) X = Al, Y = Al_2O_3 , Type of reaction = Thermal decomposition (D) X = Zn, Y = ZnO, Type of reaction = Endothermic

ANSWER: (B) X = Mg, Y = MgO, Type of reaction = Combination

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When aqueous solutions of potassium iodide and lead nitrate are mixed, an insoluble substance separates out. The chemical equation for the reaction involved is:

OPTIONS: (A) $\text{KI} + \text{PbNO}_3 \rightarrow \text{PbI} + \text{KNO}_3$ (B) $2\text{KI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbI}_2 + 2\text{KNO}_3$ (C) $\text{KI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbI} + \text{KNO}_3$ (D) $\text{KI} + \text{PbNO}_3 \rightarrow \text{PbI}_2 + \text{KNO}_3$

ANSWER: (B) $2\text{KI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbI}_2 + 2\text{KNO}_3$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When Sodium bicarbonate reacts with dilute hydrochloric acid, the gas evolved is:

OPTIONS: (A) Hydrogen; it gives pop sound with burning match stick (B) Hydrogen; it turns lime water milky (C) Carbon dioxide; it turns lime water milky (D) Carbon dioxide; it blows off a burning match stick with a pop sound

ANSWER: (C) Carbon dioxide; it turns lime water milky

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Consider the following chemical equations: I. $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$; II. $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. The correct statement about these equations is:

OPTIONS: (A) I is a displacement reaction and II is a decomposition reaction (B) I is a displacement reaction and II is a double displacement reaction (C) Both I and II are displacement reactions (D) Both I and II are double-displacement reactions

ANSWER: (B) I is a displacement reaction and II is a double displacement reaction

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The emission of brown fumes in the given experimental set-up is due to:

OPTIONS: (A) thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide (B) thermal decomposition of lead nitrate which produces brown fumes of lead oxide (C) oxidation of lead nitrate forming lead oxide and nitrogen dioxide (D) oxidation of lead nitrate forming lead oxide and oxygen

ANSWER: (A) thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The correct representation of covalent bonding in an oxygen molecule is:

OPTIONS: (A) $O=O$ (B) $O-O$ (C) $O::O$ (D) $O-O-O$

ANSWER: (A) $O=O$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not a redox reaction?

OPTIONS: (A) $2Na + 2H_2O \rightarrow 2NaOH + H_2$ (B) $NaOH + HCl \rightarrow NaCl + H_2O$ (C) $2H_2 + O_2 \rightarrow 2H_2O$ (D) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

ANSWER: (B) $NaOH + HCl \rightarrow NaCl + H_2O$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A student took a small amount of copper oxide in a conical flask and added dilute hydrochloric acid to it with constant stirring. He observed a change in colour of the solution. (i) Write the name of the compound formed and its colour. (ii) Write a balanced chemical equation for the reaction involved.

OPTIONS: (A) Copper chloride, blue-green; $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$ (B) Copper oxide, black; $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$ (C) Copper chloride, brown; $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$ (D) Copper hydroxide, blue; $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$
ANSWER: (A) Copper chloride, blue-green; $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: What is observed when aqueous solutions of potassium iodide and lead nitrate are mixed together? Name the type of reaction and write the chemical equation for the reaction that occurs.

OPTIONS: (A) Yellow precipitate of lead iodide is formed; Double displacement reaction; $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$ (B) White precipitate of lead chloride is formed; Precipitation reaction; $\text{Pb}(\text{NO}_3)_2 + 2\text{KCl} \rightarrow \text{PbCl}_2 + 2\text{KNO}_3$ (C) Yellow precipitate of lead iodide is formed; Combination reaction; $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$ (D) No reaction occurs; $\text{Pb}(\text{NO}_3)_2 + \text{KI} \rightarrow$ No reaction

ANSWER: (A) Yellow precipitate of lead iodide is formed; Double displacement reaction; $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 + 2\text{KNO}_3$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: When copper powder is heated in a watch glass, a black substance is formed. (i) Why is this black substance formed? Name it. (ii) How can this black substance be reversed to its original form?

OPTIONS: (A) Oxygen is added to copper / Copper is oxidised; copper oxide / CuO ; By passing hydrogen gas over it (B) Copper is reduced; copper oxide / CuO ; By heating it (C) Oxygen is removed from copper; copper oxide / CuO ; By passing oxygen gas over it (D) Copper reacts with nitrogen; copper nitride / Cu_3N_2 ; By passing hydrogen gas over it

ANSWER: (A) Oxygen is added to copper / Copper is oxidised; copper oxide / CuO ; By passing hydrogen gas over it

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Silver chloride kept in a china dish turns grey in sunlight. (a) Write the colour of silver chloride when it was kept in the china dish. (b) Name the type of chemical reaction taking place and write the chemical equation for the reaction. (c) State one use of the reaction. Name one more chemical which can be used for the same purpose.

OPTIONS: (A) White; Decomposition reaction / Photolytic decomposition; $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$; used in black and white photography; AgBr / Silver Bromide (B) White; Combination reaction; $2\text{Ag} + \text{Cl}_2 \rightarrow 2\text{AgCl}$; used in black and white photography; AgBr (C) Grey; Decomposition reaction; $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$; used in colour photography; AgBr (D) White; Displacement reaction; $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$; used in black and white photography; AgNO_3

ANSWER: (A) White; Decomposition reaction / Photolytic decomposition; $2\text{AgCl} \rightarrow 2\text{Ag} + \text{Cl}_2$; used in black and white photography; AgBr / Silver Bromide

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: With the help of suitable chemical equations, list the two main differences between roasting and calcination. How is metal reduced from the product obtained after roasting/calcination of the ore? Write the chemical equation for the reaction involved.

OPTIONS: (A) Roasting: for sulphide ores, heated in excess air; Calcination: for carbonate ores, heated in limited air; Reduction with carbon; $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ (B) Roasting: for carbonate ores, heated in limited air; Calcination: for sulphide ores, heated in excess air; Reduction with hydrogen; $\text{ZnO} + \text{H}_2 \rightarrow \text{Zn} + \text{H}_2\text{O}$ (C) Roasting: for sulphide ores, heated in limited air; Calcination: for carbonate ores, heated in excess air; Reduction with carbon; $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}_2$ (D) Roasting: for oxide ores, heated in excess air; Calcination: for sulphide ores, heated in limited air; Reduction with heat; $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$

ANSWER: (A) Roasting: for sulphide ores, heated in excess air; Calcination: for carbonate ores, heated in limited air; Reduction with carbon; $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Identify the reducing agent in the following reactions: $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$; $\text{H}_2\text{O} + \text{F}_2 \rightarrow \text{HF} + \text{HOF}$; $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$; $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (b) Define a redox reaction in terms of gain or loss of oxygen.

OPTIONS: (A) (i) NH_3 (ii) H_2O (iii) CO (iv) H_2 ; A reaction in which the gain or loss of oxygen takes place simultaneously is called a redox reaction (B) (i) O_2 (ii) F_2 (iii) CO (iv) O_2 ; A reaction in which gain of oxygen takes place is called a redox reaction (C) (i) NH_3 (ii) H_2O (iii) Fe_2O_3 (iv) H_2 ; A reaction in which loss of oxygen takes place is called a redox reaction (D) (i) O_2 (ii) F_2 (iii) CO (iv) O_2 ; A reaction in which gain or loss of hydrogen takes place is called a redox reaction

ANSWER: (A) (i) NH_3 (ii) H_2O (iii) CO (iv) H_2 ; A reaction in which the gain or loss of oxygen takes place simultaneously is called a redox reaction

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Write down the balanced chemical equations for the following reactions and identify the type of reaction in each case. (a) Nitrogen gas is treated with hydrogen gas to form ammonia gas. (b) Lead nitrate is heated strongly to form lead monoxide, nitrogen dioxide and oxygen. (c) A copper wire is dipped in silver nitrate solution and a shining deposit of silver is produced.

OPTIONS: (A) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (Combination); $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ (Decomposition); $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$ (Displacement) (B) $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$ (Combination); $\text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbO} + \text{NO}_2 + \text{O}_2$ (Decomposition); $\text{Cu} + \text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{Ag}$ (Displacement) (C) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (Displacement); $\text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbO} + \text{NO}_2$ (Decomposition); $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$ (Combination) (D) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (Combination); $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ (Combination); $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$ (Decomposition)

ANSWER: (A) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (Combination); $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ (Decomposition); $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$ (Displacement)

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) A saturated organic compound A belongs to the homologous series of alcohols. On heating A with concentrated sulphuric acid at 443 K, it forms an unsaturated compound B with molecular mass 28 u. The compound B on addition of one mole of hydrogen in the presence of Nickel, changes to a saturated hydrocarbon C. (i) Identify A, B and C. (ii) Write the chemical equations showing the conversion of A into B. (iii) What happens when compound C undergoes combustion? (iv) State one industrial application of hydrogenation reaction. (v) Name the products formed when compound A reacts with sodium.

OPTIONS: (A) A: Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), B: Ethene ($\text{CH}_2=\text{CH}_2$), C: Ethane (CH_3-CH_3); $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen (B) A: Methanol (CH_3OH), B: Ethene ($\text{CH}_2=\text{CH}_2$), C: Ethane (CH_3-CH_3); $\text{CH}_3\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$; C burns with sooty flame; Hydrogenation of vegetable oils; Sodium methoxide and hydrogen (C) A: Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), B: Ethyne ($\text{CH}=\text{CH}$), C: Ethane (CH_3-CH_3); $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}=\text{CH} + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and water (D) A: Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), B: Ethene ($\text{CH}_2=\text{CH}_2$), C: Ethene ($\text{CH}_2=\text{CH}_2$); $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$; C burns with sooty flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen

ANSWER: (A) A: Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), B: Ethene ($\text{CH}_2=\text{CH}_2$), C: Ethane (CH_3-CH_3); $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen

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CHAPTER: 2 - Acids, Bases and Salts (Chemistry)

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Fresh milk has a pH of 6. To delay its curdling, a chemical substance is added to it, which is:

OPTIONS: (A) Sodium carbonate (B) Baking powder (C) Sodium hydroxide (Caustic soda) (D) Baking soda (Sodium hydrogen carbonate)

ANSWER: (D) Baking soda (Sodium hydrogen carbonate)

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Hydronium ions are formed by the reaction between:

OPTIONS: (A) Sodium hydroxide and water (B) Calcium chloride and water (C)

Hydrogen chloride gas and water (D) Ethanol and water

ANSWER: (C) Hydrogen chloride gas and water

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The table below has information regarding pH and the nature of four different solutions. Which option is correct?

OPTIONS: (A) Lemon juice - pH 3 - Basic (B) Milk of magnesia - pH 10 - Basic (C)

Gastric juice - pH 6 - Acidic (D) Pure water - pH 7 - Neutral

ANSWER: (B) Milk of magnesia - pH 10 - Basic

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is true for an amphoteric oxide?

OPTIONS: (A) It reacts only with acid and does not form water (B) It reacts with acid as well as base to form salt and hydrogen gas (C) It reacts with both acid as well as base to form salt and water (D) It reacts only with base and does not form water

ANSWER: (C) It reacts with both acid as well as base to form salt and water

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Few drops of aqueous solution of ammonium chloride are put on a universal indicator paper. The paper turns pink. Study the following table and choose the correct option.

OPTIONS: (A) Acidic; weak acid and strong base; less than 7 (B) Basic; weak acid and strong base; more than 7 (C) Acidic; strong acid and weak base; less than 7 (D) Basic; strong acid and strong base; 7

ANSWER: (C) Acidic; strong acid and weak base; less than 7

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select washing soda from the following:

OPTIONS: (A) NaHCO_3 (B) $\text{Na}_2\text{CO}_3 \cdot 5\text{H}_2\text{O}$ (C) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (D) NaOH

ANSWER: (C) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The balanced chemical equation showing reaction between quicklime and water is:

OPTIONS: (A) $2\text{CaO} + \text{H}_2\text{O} \rightarrow 2\text{CaOH} + \text{H}_2 + \text{Heat}$ (B) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2 + \text{Heat}$ (C) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{Heat}$ (D) $2\text{CaO} + 3\text{H}_2\text{O} \rightarrow 2\text{Ca(OH)}_3 + \text{O}_2 + \text{Heat}$

ANSWER: (C) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{Heat}$

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following gives the correct increasing order of acidic strength?

OPTIONS: (A) Sodium chloride < Acetic acid < Hydrochloric acid (B) Sodium chloride < Hydrochloric acid < Acetic acid (C) Acetic acid < Sodium chloride < Hydrochloric acid (D) Hydrochloric acid < Sodium chloride < Acetic acid

ANSWER: (A) Sodium chloride < Acetic acid < Hydrochloric acid

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select a pair of olfactory indicators from the following:

OPTIONS: (A) Clove oil and vanilla essence (B) Onion and turmeric (C) Clove oil and litmus solution (D) Vanilla and methyl orange

ANSWER: (A) Clove oil and vanilla essence

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the experimental setup given below, it is observed that on passing the gas produced in the reaction in the solution X, the solution X first turns milky and then colourless. The option that justifies the above stated observation is that X is aqueous calcium hydroxide and:

OPTIONS: (A) it turns milky due to carbon dioxide gas liberated in the reaction and after sometime it becomes colourless due to formation of calcium carbonate (B) it turns milky due to formation of calcium carbonate and on passing excess of carbon dioxide it becomes colourless due to formation of calcium hydrogen carbonate which is soluble in water (C) it turns milky due to passing of carbon dioxide through it. It turns colourless as on further passing carbon dioxide, sodium hydrogen carbonate is formed which is soluble in water (D) the carbon dioxide liberated during the reaction turns lime water milky due to formation of calcium hydrogen carbonate and after some time it turns colourless due to formation of calcium carbonate which is soluble in water

ANSWER: (B) it turns milky due to formation of calcium carbonate and on passing excess of carbon dioxide it becomes colourless due to formation of calcium hydrogen carbonate which is soluble in water

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A solution turns the colour of turmeric to reddish brown. If the same solution is poured on universal indicator, its colour would change to:

OPTIONS: (A) violet (B) blue (C) red (D) green

ANSWER: (B) blue

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two salts X and Y are dissolved in water separately. When phenolphthalein is added to these two solutions, solution X turns pink and solution Y does not show any change in colour. Therefore X and Y are:

OPTIONS: (A) Na_2CO_3 and NH_4Cl (B) Na_2SO_4 and NaHCO_3 (C) NH_4Cl and Na_2SO_4 (D) NaNO_3 and Na_2SO_4

ANSWER: (A) Na_2CO_3 and NH_4Cl

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A compound X which is prepared from gypsum has the property of hardening when mixed with proper quantity of water. (i) Identify X and write its chemical formula. (ii) State the difference in chemical composition between baking soda and baking powder.

OPTIONS: (A) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid (B) X: Gypsum/ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid (C) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - Na_2CO_3 ; Baking Powder - a mixture of NaHCO_3 and citric acid (D) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of Na_2CO_3 and tartaric acid

ANSWER: (A) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A substance X is used as a building material and is insoluble in water. When it reacts with dil. HCl, it produces a gas which turns lime water milky. (i) Write the chemical name and formula of X. (ii) Write chemical equations for the chemical reactions involved.

OPTIONS: (A) Calcium Carbonate, CaCO_3 ; $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2\uparrow$; $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ (B) Calcium Oxide, CaO ; $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$; $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ (C) Calcium Carbonate, CaCO_3 ; $\text{CaCO}_3 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$; $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ (D) Sodium Carbonate, Na_2CO_3 ; $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$; $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
ANSWER: (A) Calcium Carbonate, CaCO_3 ; $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2\uparrow$; $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two solutions M and N give Red and Blue colour respectively with a universal indicator. (i) In which solution will the hydrogen ion concentration be more? Justify your answer. (ii) If both M and N solutions are mixed and the resultant mixture is tested with a universal indicator, it turns green. What is the nature of the salt formed? Justify your answer.

OPTIONS: (A) M (acidic) has more H^+ ions; Neutral salt is formed as acid and base neutralize each other (B) N (basic) has more H^+ ions; Acidic salt is formed (C) M (acidic) has more H^+ ions; Basic salt is formed (D) Both have equal H^+ ions; Neutral salt is formed

ANSWER: (A) M (acidic) has more H^+ ions; Neutral salt is formed as acid and base neutralize each other

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Sometimes the pH of our mouth gets lower than 5.5. Why? (b) A basic salt X is obtained by heating baking soda followed by crystallisation. Identify X and state its two industrial uses. (c) Why do copper sulphate crystals turn white on heating?

OPTIONS: (A) Bacteria produce acids by degrading sugar; X = Sodium carbonate decahydrate; Uses: Glass, soap and paper industries; Water of crystallisation is removed; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$ (B) Bacteria produce bases by degrading sugar; X = Sodium bicarbonate; Uses: Baking industry; Water of crystallisation is removed; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$ (C) Acids are produced by saliva; X = Sodium carbonate; Uses: Glass industry; Water is absorbed; $\text{CuSO}_4 \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (D) Bacteria produce acids by degrading sugar; X = Sodium carbonate decahydrate; Uses: Food preservation; Water of crystallisation is removed; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + \text{H}_2\text{O}$

ANSWER: (A) Bacteria produce acids by degrading sugar; X = Sodium carbonate decahydrate; Uses: Glass, soap and paper industries; Water of crystallisation is removed; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (i) Suggest a safe procedure of diluting a strong concentrated acid. (ii) Name the salt formed when sulphuric acid is added to sodium hydroxide and write its pH. (iii) Dry HCl gas does not change the colour of dry blue litmus paper. Why?

OPTIONS: (A) Add acid slowly to water with constant stirring; Sodium sulphate/ Na_2SO_4 , pH = 7; Dry HCl is unable to ionise (B) Add water to acid slowly with constant stirring; Sodium sulphate/ Na_2SO_4 , pH = 7; Dry HCl is unable to ionise (C) Add acid slowly to water with constant stirring; Sodium chloride/ NaCl , pH = 7; Dry HCl is unable to ionise (D) Add acid slowly to water with constant stirring; Sodium sulphate/ Na_2SO_4 , pH = 5; Dry HCl is unable to ionise

ANSWER: (A) Add acid slowly to water with constant stirring; Sodium sulphate/ Na_2SO_4 , pH = 7; Dry HCl is unable to ionise

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chemical compound X is used in glass, soap and paper industries. On treatment with ethanoic acid, it forms salt, water and carbon dioxide. (a) Identify X. (b) How is this compound obtained from brine? Write the equations involved. (c) State the number of molecules of water of crystallisation present in compound X.

OPTIONS: (A) $X = \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (Washing soda); $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$; $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$; $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; 10 molecules of water (B) $X = \text{NaHCO}_3$ (Baking soda); $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$; $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$; 10 molecules of water (C) $X = \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (Washing soda); $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$; $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; 5 molecules of water (D) $X = \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (Washing soda); $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$; $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$; 10 molecules of water

ANSWER: (A) $X = \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (Washing soda); $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$; $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$; $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$; 10 molecules of water

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Consider the following salts: (i) YCl (ii) NH_4X (iii) ZCO_3 (a) What would be the pH of the salt solution if in YCl , Y is sodium? Give reason for your answer. (b) If in salt NH_4X , X is nitrate, then its solution will give what colour with universal indicator? Why? (c) What would be the change in colour in blue litmus solution if ZCO_3 is added to it and Z is potassium?

OPTIONS: (A) $\text{pH} = 7$ (salt of strong acid and strong base); Pink/Orange (salt of weak base and strong acid); No change/remains blue (B) $\text{pH} > 7$ (salt of strong acid and strong base); Pink/Orange (salt of weak base and strong acid); Turns red (C) $\text{pH} = 7$ (salt of weak acid and weak base); Blue (salt of strong base and strong acid); No change (D) $\text{pH} = 7$ (salt of strong acid and strong base); Green (salt of weak base and strong acid); Turns red

ANSWER: (A) $\text{pH} = 7$ (salt of strong acid and strong base); Pink/Orange (salt of weak base and strong acid); No change/remains blue

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The teacher conducted a practical to find pH of various solutions. Group A: Lemon juice, vinegar, colourless aerated drink. Group B: Tomato juice, coffee, ginger juice. Group C: Sodium hydroxide, sodium chloride, lime water. (a) For the solutions provided, which group is/are likely to have pH value (i) less than 7, and (ii) greater than 7? (b) List two ways of determining pH of a solution. (c) Explain why sour substances such as lemon juice are effective in cleaning tarnished copper vessels.

OPTIONS: (A) (i) Groups A and B - less than 7; (ii) Group C - greater than 7; pH paper and universal indicator; Copper vessel is tarnished due to formation of basic copper oxide. Lemon juice being acidic reacts with copper oxide and the salt formed is washed away (B) (i) Group A - less than 7; (ii) Groups B and C - greater than 7; pH paper and litmus paper; Copper vessel is tarnished due to formation of acidic copper oxide (C) (i) Groups A and C - less than 7; (ii) Group B - greater than 7; pH paper and universal indicator; Lemon juice being acidic dissolves copper (D) (i) Groups A and B - less than 7; (ii) Group C - greater than 7; pH paper and methyl orange; Copper vessel is tarnished due to formation of copper chloride

ANSWER: (A) (i) Groups A and B - less than 7; (ii) Group C - greater than 7; pH paper and universal indicator; Copper vessel is tarnished due to formation of basic copper oxide. Lemon juice being acidic reacts with copper oxide and the salt formed is washed away

YEAR: 2023

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) An acid X and an alcohol Y react with each other in the presence of an acid catalyst to form a sweet smelling substance Z. Identify X, Y and Z. Write the chemical equation for the reaction involved and name it. The substance Z on treatment with sodium hydroxide produces back the alcohol Y and sodium ethanoate. Write the chemical equation for the reaction involved and name it, giving justification for the name.

OPTIONS: (A) X = CH₃COOH (Ethanoic Acid), Y = C₂H₅OH (Ethanol), Z = CH₃COOC₂H₅ (Ethyl ethanoate); CH₃COOH + C₂H₅OH → CH₃COOC₂H₅ + H₂O (Esterification); CH₃COOC₂H₅ + NaOH → C₂H₅OH + CH₃COONa (Saponification); It is used in preparation of soap (B) X = CH₃COOH (Ethanoic Acid), Y = CH₃OH (Methanol), Z = CH₃COOCH₃ (Methyl ethanoate); CH₃COOH + CH₃OH → CH₃COOCH₃ + H₂O (Esterification); CH₃COOCH₃ + NaOH → CH₃OH + CH₃COONa (Saponification); It is used in preparation of soap (C) X = HCOOH (Methanoic Acid), Y = C₂H₅OH (Ethanol), Z = HCOOC₂H₅ (Ethyl methanoate); HCOOH + C₂H₅OH → HCOOC₂H₅ + H₂O (Esterification); HCOOC₂H₅ + NaOH → C₂H₅OH + HCOONa (Saponification); It is used in preparation of soap (D) X = CH₃COOH (Ethanoic Acid), Y = C₂H₅OH (Ethanol), Z = CH₃COOC₂H₅ (Ethyl ethanoate); CH₃COOH + C₂H₅OH → CH₃COOC₂H₅ + H₂O (Esterification); CH₃COOC₂H₅ + NaOH → C₂H₅OH + CH₃COONa (Neutralization); It is used in preparation of soap

ANSWER: (A) X = CH₃COOH (Ethanoic Acid), Y = C₂H₅OH (Ethanol), Z = CH₃COOC₂H₅ (Ethyl ethanoate); CH₃COOH + C₂H₅OH → CH₃COOC₂H₅ + H₂O (Esterification); CH₃COOC₂H₅ + NaOH → C₂H₅OH + CH₃COONa (Saponification); It is used in preparation of soap

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CHAPTER: 3 - Metals and Non-metals (Chemistry)

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YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Metal oxides generally react with acids, but few oxides of metal also react with bases. Such metallic oxides are:

OPTIONS: (A) MgO and ZnO (B) ZnO and Al₂O₃ (C) Al₂O₃ and CaO (D) MgO and CaO

ANSWER: (B) ZnO and Al₂O₃

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A metal X is used in thermite process. When X is burnt in air it gives an amphoteric oxide Y. X and Y are respectively:

OPTIONS: (A) Fe and Fe_2O_3 (B) Al and Al_2O_3 (C) Fe and Fe_3O_4 (D) Al and Al_3O_4

ANSWER: (B) Al and Al_2O_3

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Copper is used for making cooking utensils. Which of the following physical properties of copper is NOT responsible for the same?

OPTIONS: (A) Malleability (B) High melting point (C) Thermal conductivity (D) High reactivity

ANSWER: (D) High reactivity

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Among the following, the metal with lowest density is:

OPTIONS: (A) Lithium (B) Lead (C) Magnesium (D) Aluminium

ANSWER: (A) Lithium

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following metals do not corrode in moist air?

OPTIONS: (A) Copper (B) Iron (C) Gold (D) Silver

ANSWER: (C) Gold

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of electrons in the outermost shell of the atom of a non-metal can be:

OPTIONS: (A) 1, 2 or 3 (B) 3, 4 or 5 (C) 5, 6 or 7 (D) 5, 6 or 8

ANSWER: (C) 5, 6 or 7

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Bronze is an alloy of:

OPTIONS: (A) Copper and Zinc (B) Aluminium and Tin (C) Copper, Tin and Zinc (D) Copper and Tin

ANSWER: (D) Copper and Tin

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The following diagram shows the electrolytic refining of copper. Which of the following statements is incorrect description of the process?

OPTIONS: (A) The impure metal from the anode dissolves into the electrolyte (B) The pure metal from the electrolyte is deposited on the cathode (C) Insoluble impurities settle down at the bottom of the anode (D) On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte

ANSWER: (D) On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The electron dot structure of chlorine molecule is:

OPTIONS: (A) Cl:Cl (B) Cl-Cl (C) Cl::Cl (D) Cl-Cl-Cl

ANSWER: (A) Cl:Cl

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total number of electrons shared in the formation of an ethyne molecule is:

OPTIONS: (A) 6 (B) 3 (C) 10 (D) 4

ANSWER: (A) 6

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Sodium hydroxide is termed an alkali while Ferric hydroxide is not because:

OPTIONS: (A) Sodium hydroxide is a strong base, while Ferric hydroxide is a weak base (B) Sodium hydroxide is a base which is soluble in water while Ferric hydroxide is also a base but it is not soluble in water (C) Sodium hydroxide is a strong base while Ferric hydroxide is a strong acid (D) Sodium hydroxide and Ferric hydroxide both are strong base but the solubility of Sodium hydroxide in water is comparatively higher than that of Ferric hydroxide

ANSWER: (B) Sodium hydroxide is a base which is soluble in water while Ferric hydroxide is also a base but it is not soluble in water

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The name of the salt used to remove permanent hardness of water is:

OPTIONS: (A) Sodium hydrogen carbonate (NaHCO_3) (B) Sodium chloride (NaCl) (C) Sodium carbonate decahydrate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) (D) Calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$)

ANSWER: (C) Sodium carbonate decahydrate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$)

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A metal nitrate A on heating gives a metal oxide along with evolution of a brown coloured gas B and a colourless gas, which helps in burning. Aqueous solution of A when reacted with potassium iodide forms a yellow precipitate. (a) Identify A and B (b) Name the types of both the reactions involved.

OPTIONS: (A) A = Lead Nitrate/ $\text{Pb}(\text{NO}_3)_2$, B = Nitrogen dioxide/ NO_2 ; Decomposition reaction, Double-displacement/Precipitation reaction (B) A = Silver Nitrate/ AgNO_3 , B = Nitrogen dioxide/ NO_2 ; Decomposition reaction, Double-displacement reaction (C) A = Lead Nitrate/ $\text{Pb}(\text{NO}_3)_2$, B = Oxygen/ O_2 ; Decomposition reaction, Precipitation reaction (D) A = Lead Nitrate/ $\text{Pb}(\text{NO}_3)_2$, B = Nitrogen dioxide/ NO_2 ; Combination reaction, Double-displacement reaction

ANSWER: (A) A = Lead Nitrate/ $\text{Pb}(\text{NO}_3)_2$, B = Nitrogen dioxide/ NO_2 ; Decomposition reaction, Double-displacement/Precipitation reaction

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Write the constituents of solder alloy. Which property of solder makes it suitable for welding electrical wires?

OPTIONS: (A) Lead and Tin ($\text{Pb}+\text{Sn}$); Low melting point (B) Copper and Zinc ($\text{Cu}+\text{Zn}$); High melting point (C) Lead and Tin ($\text{Pb}+\text{Sn}$); High melting point (D) Copper and Tin ($\text{Cu}+\text{Sn}$); Low melting point

ANSWER: (A) Lead and Tin ($\text{Pb}+\text{Sn}$); Low melting point

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) What happens when copper is heated in air? (Give the equation of the reaction involved). (b) Why are some metal oxides categorized as amphoteric? Give one example. (c) Complete the following equations: $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow$

OPTIONS: (A) $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$; Because they react with both acids and bases to produce salt and water; $\text{Al}_2\text{O}_3/\text{ZnO}$; $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq})$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ (B) $\text{Cu} + \text{O}_2 \rightarrow \text{CuO}$; Because they react only with acids; Al_2O_3 ; $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NaOH}$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow \text{NaAlO}_2 + \text{H}_2\text{O}$ (C) $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$; Because they react only with bases; ZnO ; $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2$ (D) $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$; Because they react with both acids and bases to produce salt and hydrogen; Al_2O_3 ; $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq})$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2$

ANSWER: (A) $2\text{Cu} + \text{O}_2 \rightarrow 2\text{CuO}$; Because they react with both acids and bases to produce salt and water; $\text{Al}_2\text{O}_3/\text{ZnO}$; $\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq})$; $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Write a balanced equation to show the reaction that occurs when a piece of aluminium is dipped in a dilute solution of (i) sulphuric acid and (ii) sodium hydroxide. (b) When copper oxide reacts with hydrochloric acid, mention the colour of the solution formed. Give reason for this observation.

OPTIONS: (A) (i) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$; (ii) $2\text{Al} + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2$; Blue-green; due to formation of Copper chloride/ CuCl_2 (B) (i) $\text{Al} + \text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$; (ii) $\text{Al} + \text{NaOH} \rightarrow \text{NaAlO}_2 + \text{H}_2$; Colourless; due to formation of Aluminium chloride (C) (i) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$; (ii) $2\text{Al} + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2$; Blue-green; due to formation of Aluminium chloride (D) (i) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$; (ii) $\text{Al} + \text{NaOH} \rightarrow \text{NaAlO}_2 + \text{H}_2\text{O}$; Blue-green; due to formation of Copper chloride

ANSWER: (A) (i) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$; (ii) $2\text{Al} + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2$; Blue-green; due to formation of Copper chloride/ CuCl_2

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: On burning Sulphur in oxygen a colourless gas is produced. (i) Write chemical equation for the reaction. (ii) Name the gas formed. (iii) State the nature of the gas. (iv) What will be the action of this on a dry litmus paper?

OPTIONS: (A) $S + O_2 \rightarrow SO_2$; Sulphur dioxide; Acidic; No change (B) $S + O_2 \rightarrow SO_2$; Sulphur dioxide; Basic; Turns blue litmus red (C) $S + O_2 \rightarrow SO_3$; Sulphur trioxide; Acidic; No change (D) $S + O_2 \rightarrow SO_2$; Sulphur dioxide; Acidic; Turns blue litmus red

ANSWER: (A) $S + O_2 \rightarrow SO_2$; Sulphur dioxide; Acidic; No change

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A reddish brown metal used in electrical wires when powdered and heated strongly turns black. When hydrogen gas is passed over this black substance, it regains its original colour. (a) Name the metal and the black substance formed. (b) Write balanced chemical equations for the two reactions involved.

OPTIONS: (A) Cu / Copper; CuO / Copper Oxide; $2Cu + O_2 \rightarrow 2CuO$; $CuO + H_2 \rightarrow Cu + H_2O$ (B) Cu / Copper; CuO₂ / Copper dioxide; $2Cu + O_2 \rightarrow 2CuO$; $CuO + H_2 \rightarrow Cu + H_2O$ (C) Fe / Iron; FeO / Iron oxide; $2Fe + O_2 \rightarrow 2FeO$; $FeO + H_2 \rightarrow Fe + H_2O$ (D) Cu / Copper; CuO / Copper Oxide; $Cu + O_2 \rightarrow CuO$; $CuO + H_2 \rightarrow Cu + H_2O$

ANSWER: (A) Cu / Copper; CuO / Copper Oxide; $2Cu + O_2 \rightarrow 2CuO$; $CuO + H_2 \rightarrow Cu + H_2O$

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) What happens when zinc carbonate is heated strongly in a limited supply of air? (b) The reaction of a metal A with Fe₂O₃ is highly exothermic and is used to join railway tracks. (I) Identify the metal A and name the reaction taking place. (II) Write the chemical equation for the reaction of metal A with Fe₂O₃.

OPTIONS: (A) It forms Zinc oxide (Calcination); $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$; A = Aluminium, Thermit Reaction; $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$ (B) It forms Zinc oxide (Roasting); $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$; A = Magnesium, Thermite Reaction; $\text{Fe}_2\text{O}_3 + 2\text{Mg} \rightarrow 2\text{Fe} + \text{Mg}_2\text{O}_3 + \text{Heat}$ (C) It forms Zinc metal; $\text{ZnCO}_3 \rightarrow \text{Zn} + \text{CO}_2$; A = Aluminium, Thermit Reaction; $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$ (D) It forms Zinc oxide (Calcination); $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}$; A = Aluminium, Thermit Reaction; $\text{Fe}_2\text{O}_3 + \text{Al} \rightarrow \text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$

ANSWER: (A) It forms Zinc oxide (Calcination); $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$; A = Aluminium, Thermit Reaction; $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Name the process of reduction used for a metal that gives vigorous reaction with air and water both. (b) Carbon cannot be used as a reducing agent to obtain aluminium from its oxide. Why? (c) Describe briefly the method to obtain mercury from cinnabar. Write the chemical equation for the reactions involved in the process.

OPTIONS: (A) Electrolytic reduction; Carbon cannot reduce oxides of highly reactive metals / these metals have more affinity for oxygen than carbon; Cinnabar heated in air $\rightarrow \text{HgO}$; $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$; $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ (B) Thermal reduction; Carbon cannot reduce oxides of highly reactive metals; Cinnabar heated in air $\rightarrow \text{HgS}$; $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$; $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ (C) Electrolytic reduction; Carbon cannot reduce oxides because it is less reactive; Cinnabar heated in air $\rightarrow \text{HgO}$; $\text{HgS} + \text{O}_2 \rightarrow \text{HgO} + \text{SO}_2$; $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$ (D) Chemical reduction; Carbon cannot reduce oxides of highly reactive metals; Cinnabar heated in air $\rightarrow \text{HgO}$; $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$; $\text{HgO} \rightarrow \text{Hg} + \text{O}_2$

ANSWER: (A) Electrolytic reduction; Carbon cannot reduce oxides of highly reactive metals / these metals have more affinity for oxygen than carbon; Cinnabar heated in air $\rightarrow \text{HgO}$; $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$; $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$

YEAR: 2023

CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (i) A metal M on reacting with dilute acid liberates a gas G. The same metal also liberates gas G when reacts with a base. (a) Write the name of gas G. (b) How will you test the presence of this gas? (c) Write chemical equations for the reactions of the metal with (1) an acid and (2) a base. (ii) We cannot use carbon to obtain sodium from sodium oxide. Why? State the reactions taking place at cathode and anode during electrolytic reduction of sodium chloride.

OPTIONS: (A) (i) Hydrogen/H₂; Gas burns with pop sound; $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$; $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$; Sodium has more affinity for oxygen than carbon; At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; At anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (B) (i) Oxygen/O₂; Gas supports combustion; $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$; $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$; Carbon has more affinity for oxygen than sodium; At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; At anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (C) (i) Hydrogen/H₂; Gas burns with pop sound; $\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$; $\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$; Sodium has more affinity for oxygen than carbon; At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; At anode: $\text{Cl}^- \rightarrow \text{Cl}_2 + \text{e}^-$ (D) (i) Hydrogen/H₂; Gas extinguishes flame; $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$; $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$; Sodium has less affinity for oxygen than carbon; At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; At anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

ANSWER: (A) (i) Hydrogen/H₂; Gas burns with pop sound; $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$; $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$; Sodium has more affinity for oxygen than carbon; At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; At anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

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CHAPTER: 4 - Carbon and its Compounds (Chemistry)

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YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Consider the structures of the three cyclic carbon compounds A, B and C given below and select the correct option:

OPTIONS: (A) A and C are isomers of hexane and B is benzene (B) A is an isomer of hexane, B is benzene and C is an isomer of hexene (C) A is a saturated cyclic hydrocarbon and B and C are unsaturated cyclic hydrocarbons (D) A is cyclohexane and B and C are the isomers of benzene

ANSWER: (C) A is a saturated cyclic hydrocarbon and B and C are unsaturated cyclic hydrocarbons

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When ethanol reacts with sodium two products are formed. These products are:

OPTIONS: (A) Sodium ethanoate and oxygen (B) Sodium ethanoate and hydrogen (C) Sodium ethoxide and oxygen (D) Sodium ethoxide and hydrogen

ANSWER: (D) Sodium ethoxide and hydrogen

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not an unsaturated hydrocarbon?

OPTIONS: (A) Ethene (B) Propene (C) Butane (D) Butyne

ANSWER: (C) Butane

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A compound X which is prepared from gypsum has the property of hardening when mixed with proper quantity of water. (i) Identify X and write its chemical formula. (ii) State the difference in chemical composition between baking soda and baking powder.

OPTIONS: (A) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid (B) X: Gypsum/ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid (C) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - Na_2CO_3 ; Baking Powder - a mixture of NaHCO_3 and citric acid (D) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of Na_2CO_3 and tartaric acid

ANSWER: (A) X: Plaster of Paris/ $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$; Baking Soda - NaHCO_3 ; Baking Powder - a mixture of NaHCO_3 and tartaric acid

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) A saturated organic compound A belongs to the homologous series of alcohols. On heating A with concentrated sulphuric acid at 443 K, it forms an unsaturated compound B with molecular mass 28 u. The compound B on addition of one mole of hydrogen in the presence of Nickel, changes to a saturated hydrocarbon C. (i) Identify A, B and C. (ii) Write the chemical equations showing the conversion of A into B. (iii) What happens when compound C undergoes combustion? (iv) State one industrial application of hydrogenation reaction. (v) Name the products formed when compound A reacts with sodium.

OPTIONS: (A) A: Ethanol, B: Ethene, C: Ethane; $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen (B) A: Methanol, B: Ethene, C: Ethane; $\text{CH}_3\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; C burns with sooty flame; Hydrogenation of vegetable oils; Sodium methoxide and hydrogen (C) A: Ethanol, B: Ethyne, C: Ethane; $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_2 + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and water (D) A: Ethanol, B: Ethene, C: Ethene; $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; C burns with sooty flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen

ANSWER: (A) A: Ethanol, B: Ethene, C: Ethane; $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; C burns with blue flame; Hydrogenation of vegetable oils; Sodium ethoxide and hydrogen

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (b) (i) Name the simplest saturated hydrocarbon. Draw its electron dot structure. Which type of bonds exist in this compound? (ii) Name any two mixtures of the carbon compound used as a fuel in daily life, of which the above mentioned compound is an important component. (iii) In which homologous series of carbon compounds can this compound be placed? Write the general formula of the series. (iv) Which type of flame is produced on burning it?

OPTIONS: (A) Methane/ CH_4 ; Covalent bond/Single bond; Biogas, CNG; Alkanes, $\text{C}_n\text{H}_{2n+2}$; Clean flame/blue flame (B) Ethane/ C_2H_6 ; Covalent bond/Double bond; LPG, CNG; Alkenes, C_nH_{2n} ; Blue flame (C) Methane/ CH_4 ; Covalent bond/Single bond;

Petrol, Diesel; Alkanes, C_nH_{2n+2} ; Sooty flame (D) Methane/ CH_4 ; Ionic bond; Biogas, CNG; Alkanes, C_nH_{2n+2} ; Clean flame

ANSWER: (A) Methane/ CH_4 ; Covalent bond/Single bond; Biogas, CNG; Alkanes, C_nH_{2n+2} ; Clean flame/blue flame

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) A neutral organic compound X (Molecular formula C_2H_6O) on reacting with acidified $K_2Cr_2O_7$ gives an organic compound Y which is acidic in nature. X reacts with Y on warming in the presence of conc. H_2SO_4 to give a sweet smelling compound Z. (i) Identify X, Y and Z. (ii) Write the chemical equations for the reactions in the conversion of (1) X to Y and (2) X to Z. (iii) State the role of (1) acidified $K_2Cr_2O_7$ in the conversion of X to Y and (2) conc. H_2SO_4 in the reaction of X and Y. (iv) Name the reaction which occurs when Z reacts with an alkali.

OPTIONS: (A) X: Ethanol, Y: Ethanoic acid/Acetic acid, Z: Ethyl ethanoate; $C_2H_5OH \rightarrow CH_3COOH$; $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$; Oxidising agent; Dehydrating agent; Saponification reaction (B) X: Methanol, Y: Methanoic acid, Z: Methyl methanoate; $CH_3OH \rightarrow HCOOH$; $HCOOH + CH_3OH \rightarrow HCOOCH_3 + H_2O$; Oxidising agent; Dehydrating agent; Saponification reaction (C) X: Ethanol, Y: Ethanoic acid, Z: Ethyl ethanoate; $C_2H_5OH \rightarrow CH_3COOH$; $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$; Reducing agent; Dehydrating agent; Esterification reaction (D) X: Ethanol, Y: Ethanoic acid, Z: Ethyl ethanoate; $C_2H_5OH \rightarrow CH_3COOH$; $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$; Oxidising agent; Catalyst; Saponification reaction
ANSWER: (A) X: Ethanol, Y: Ethanoic acid/Acetic acid, Z: Ethyl ethanoate; $C_2H_5OH \rightarrow CH_3COOH$; $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$; Oxidising agent; Dehydrating agent; Saponification reaction

YEAR: 2023

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (b) (i) What are isomers? Write the structures of two compounds having molecular formula C_3H_6O and give their names. (ii) What are soaps? How are they chemically different from detergents? Why do soaps not work effectively in hard water?

OPTIONS: (A) Compounds having same molecular formula but different structural formula; Propanone and Propanal; Soaps are sodium or potassium salts of long-chain carboxylic acids; Detergents are sodium salts of sulphonic acids; Soaps react with calcium or magnesium ions in hard water to form scum (B) Compounds having different molecular formula but same structure; Propanone and Propanol; Soaps are sodium salts of long-chain carboxylic acids; Detergents are sodium salts of sulphonic acids; Soaps react with sodium ions in hard water (C) Compounds having same molecular formula but different structure; Propanal and Propanol; Soaps are potassium salts of long-chain carboxylic acids; Detergents are sodium salts of hydrochloric acid; Soaps react with calcium or magnesium ions in hard water to form scum (D) Compounds having same molecular formula but different structure; Propanone and Propanal; Soaps are sodium or potassium salts of long-chain carboxylic acids; Detergents are sodium salts of sulphonic acids; Soaps do not react with hard water

ANSWER: (A) Compounds having same molecular formula but different structural formula; Propanone and Propanal; Soaps are sodium or potassium salts of long-chain carboxylic acids; Detergents are sodium salts of sulphonic acids; Soaps react with calcium or magnesium ions in hard water to form scum

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CHAPTER: 5 - Life Processes (Biology)

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YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An organism which breaks down the food material outside the body and then absorbs it is:

OPTIONS: (A) a plant parasite, Cuscuta (B) an animal parasite, Tapeworm (C) a bacteria, Rhizobium (D) a fungi, Rhizopus

ANSWER: (D) a fungi, Rhizopus

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Consider the following statements about small intestine and select the one which is NOT correct:

OPTIONS: (A) The length of the small intestine in animals differs as it depends on the type of food they eat (B) The small intestine is the site of complete digestion of food (C) The small intestine receives secretions from liver and pancreas (D) The villi of the small intestine absorb water from the unabsorbed food before it gets removed from the body via the anus

ANSWER: (D) The villi of the small intestine absorb water from the unabsorbed food before it gets removed from the body via the anus

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: During vigorous exercise, the occurrence of cramps in the outer muscles of an athlete is due to the conversion of pyruvate to:

OPTIONS: (A) Glucose (B) Ethanol (C) Lactic acid (D) Lactose

ANSWER: (C) Lactic acid

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the following diagram, identify the cells through which massive amounts of gaseous exchange takes place for photosynthesis:

OPTIONS: (A) I (B) IV (C) III (D) II

ANSWER: (B) IV

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Sphincter muscles are present at the exit of:

OPTIONS: (A) Stomach and small intestine (B) Stomach and anus (C) Small intestine and large intestine (D) Oesophagus and stomach

ANSWER: (B) Stomach and anus

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Opening and closing of stomata is due to:

OPTIONS: (A) High pressure of gases inside the cells (B) Movement of water in and out of the guard cells (C) Stimulus of light in the guard cells (D) Diffusion of CO₂ in and out of the guard cells

ANSWER: (B) Movement of water in and out of the guard cells

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Water in the root enters due to:

OPTIONS: (A) the function of the root to absorb water (B) difference in the concentration of ions between the root and the soil (C) excess water present in the soil (D) diffusion of water in the roots

ANSWER: (B) difference in the concentration of ions between the root and the soil

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: As compared to terrestrial organisms, the rate of breathing in aquatic organisms is:

OPTIONS: (A) faster because they need more oxygen for their survival (B) faster because the amount of dissolved oxygen in water is fairly low (C) slower because the amount of dissolved oxygen in water is fairly low (D) slower because the capacity of water of dissolving atmospheric air is limited

ANSWER: (B) faster because the amount of dissolved oxygen in water is fairly low

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The process in which loss of water in the form of vapours from the aerial parts of plants takes place is X, which helps in Y. Here X and Y respectively are:

OPTIONS: (A) transpiration and photosynthesis (B) transpiration and temperature regulation (C) translocation and movement of soluble products of photosynthesis in phloem (D) translocation and absorption of water and minerals from soil by roots

ANSWER: (B) transpiration and temperature regulation

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not a function of the human kidney?

OPTIONS: (A) Filtering blood to remove nitrogenous wastes (B) Regulating water balance in the body (C) Producing digestive enzymes (D) Maintaining pH balance of blood

ANSWER: (C) Producing digestive enzymes

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The enzyme that breaks down proteins in the stomach is:

OPTIONS: (A) Amylase (B) Lipase (C) Pepsin (D) Trypsin

ANSWER: (C) Pepsin

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The process by which plants lose water in the form of water vapour is called:

OPTIONS: (A) Translocation (B) Transpiration (C) Photosynthesis (D) Respiration

ANSWER: (B) Transpiration

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The blood vessel that carries oxygenated blood from the lungs to the heart is:

OPTIONS: (A) Pulmonary artery (B) Pulmonary vein (C) Aorta (D) Vena cava

ANSWER: (B) Pulmonary vein

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Name the part of the human excretory system where nephrons are found. Write the structure and function of nephrons.

OPTIONS: (A) Kidneys; Structure: A cluster of thin-walled capillaries (glomerulus) associated with cup-shaped end of a tube called Bowman's capsule; Function: Filtration of nitrogenous waste from blood to form urine / Reabsorption of useful materials (B) Liver; Structure: A cluster of thin-walled capillaries; Function: Filtration of blood (C) Kidney; Structure: A cluster of thick-walled capillaries; Function: Reabsorption of water only (D) Bladder; Structure: A cluster of thin-walled capillaries; Function: Storage of urine

ANSWER: (A) Kidneys; Structure: A cluster of thin-walled capillaries (glomerulus) associated with cup-shaped end of a tube called Bowman's capsule; Function: Filtration of nitrogenous waste from blood to form urine / Reabsorption of useful materials

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Write the role of insulin in regulating blood sugar levels in human body. Mention the disease caused due to it. How is the timing and the amount of release of insulin in the blood regulated?

OPTIONS: (A) Lowers blood sugar levels; Diabetes; The rise in sugar level in blood produces more insulin. As the blood sugar level falls, secretion is reduced (B) Raises blood sugar levels; Hyperglycemia; The fall in sugar level in blood produces less insulin. As the blood sugar level rises, secretion increases (C) Converts sugar to glycogen; Diabetes; The rise in sugar level in blood produces less insulin. As the blood sugar level falls, secretion increases (D) Lowers blood sugar levels; Hypoglycemia; The rise in sugar level in blood produces more insulin. As the blood sugar level falls, secretion is reduced

ANSWER: (A) Lowers blood sugar levels; Diabetes; The rise in sugar level in blood produces more insulin. As the blood sugar level falls, secretion is reduced

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Name the type of blood (oxygenated/deoxygenated) transported by each of the following mentioning the path: (i) Vena cava (ii) Pulmonary artery

OPTIONS: (A) Vena cava - deoxygenated blood from body to heart; Pulmonary artery - deoxygenated blood from heart to lungs (B) Vena cava - oxygenated blood from body to heart; Pulmonary artery - oxygenated blood from heart to lungs (C) Vena cava - deoxygenated blood from heart to body; Pulmonary artery - oxygenated blood from lungs to heart (D) Vena cava - oxygenated blood from heart to body; Pulmonary artery - deoxygenated blood from heart to lungs

ANSWER: (A) Vena cava - deoxygenated blood from body to heart; Pulmonary artery - deoxygenated blood from heart to lungs

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) (i) What is the first step in the breakdown of glucose during aerobic and anaerobic respiration? Where does it take place? (ii) ATP is called the energy currency of the cell. Why? (iii) What is meant by residual volume of air in a breathing cycle?

OPTIONS: (A) Glucose → Pyruvate (in cytoplasm); ATP is used as fuel for all activities in a cell / ATP is broken down giving rise to a fixed amount of energy; Volume of air present in lung after exhalation (B) Glucose → Lactic acid (in mitochondria); ATP is stored energy; Volume of air present in lung after inhalation (C) Glucose → Pyruvate (in mitochondria); ATP is used for photosynthesis; Volume of air present in lung after exhalation (D) Glucose → Carbon dioxide (in cytoplasm); ATP is used as fuel for all activities in a cell; Volume of air present in lung after inhalation

ANSWER: (A) Glucose → Pyruvate (in cytoplasm); ATP is used as fuel for all activities in a cell / ATP is broken down giving rise to a fixed amount of energy; Volume of air present in lung after exhalation

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Write the sequence of events that involve response of a person when a dust particle is inhaled through the nose by him.

OPTIONS: (A) Dust particles are trapped by hair and mucus in the nasal passage; This triggers sneezing reflex to expel the particles (B) Dust particles pass through the trachea; They are absorbed by the blood; This triggers coughing (C) Dust particles reach the lungs; They are absorbed by the alveoli; This triggers asthma (D) Dust particles are trapped by the cilia in the trachea; This triggers coughing

ANSWER: (A) Dust particles are trapped by hair and mucus in the nasal passage; This triggers sneezing reflex to expel the particles

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) (i) How does Paramecium obtain its food? (ii) List the role of each of the following in our digestive system: (a) Hydrochloric acid (b) Trypsin (c) Muscular walls of stomach (d) Salivary amylase

OPTIONS: (A) Food enters through a specific spot with the help of movement of cilia; HCl - creates acidic medium for enzyme action/kills microbes; Trypsin - digestion of proteins; Muscular walls - mixing food with digestive juices/pushes food; Salivary amylase - conversion of starch into sugar (B) Food is engulfed through pseudopodia; HCl - digestion of fats; Trypsin - digestion of carbohydrates; Muscular walls - absorption of food; Salivary amylase - conversion of proteins into amino acids (C) Food enters through a specific spot with the help of movement of cilia; HCl - digestion of proteins; Trypsin - digestion of fats; Muscular walls - storage of food; Salivary amylase - conversion of sugar into starch (D) Food is engulfed through pseudopodia; HCl - creates alkaline medium; Trypsin - digestion of proteins; Muscular walls - mixing food; Salivary amylase - conversion of starch into sugar

ANSWER: (A) Food enters through a specific spot with the help of movement of cilia; HCl - creates acidic medium for enzyme action/kills microbes; Trypsin - digestion of proteins; Muscular walls - mixing food with digestive juices/pushes food; Salivary amylase - conversion of starch into sugar

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two green plants are kept separately in oxygen free containers, one in the dark and other in sunlight. It was observed that plant kept in dark could not survive longer. Give reason for this observation.

OPTIONS: (A) The plant kept in dark is unable to carry out photosynthesis and due to absence of oxygen it cannot respire. But the plant kept in light is able to photosynthesize converting CO₂ into oxygen which it can use for respiration (B) The plant kept in dark cannot carry out respiration and due to absence of CO₂ it cannot photosynthesize (C) The plant kept in dark cannot carry out transpiration and due to absence of water it cannot survive (D) The plant kept in dark cannot carry out photosynthesis but it can respire using stored food

ANSWER: (A) The plant kept in dark is unable to carry out photosynthesis and due to absence of oxygen it cannot respire. But the plant kept in light is able to photosynthesize converting CO₂ into oxygen which it can use for respiration

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: With the help of an activity, explain the action of saliva on the food we eat. Why is bile juice important in the process of digestion?

OPTIONS: (A) Activity: Take 1 ml starch solution in two test tubes, add saliva to one, leave for 20-30 minutes, add iodine. The colour of iodine does not change in test tube with saliva as starch is converted to sugar. Bile juice makes food alkaline for pancreatic enzymes and emulsifies fats (B) Activity: Take 1 ml sugar solution in two test tubes, add saliva to one, leave for 20-30 minutes, add iodine. The colour changes in test tube with saliva as sugar is converted to starch. Bile juice digests proteins (C) Activity: Take 1 ml starch solution in two test tubes, add saliva to one, leave for 20-30 minutes, add iodine. The colour changes to blue-black in test tube with saliva as starch is formed. Bile juice makes food acidic for stomach enzymes (D) Activity: Take 1 ml protein solution in two test tubes, add saliva to one, leave for 20-30 minutes, add iodine. The colour does not change in test tube with saliva as proteins are digested. Bile juice emulsifies fats

ANSWER: (A) Activity: Take 1 ml starch solution in two test tubes, add saliva to one, leave for 20-30 minutes, add iodine. The colour of iodine does not change in test tube with saliva as starch is converted to sugar. Bile juice makes food alkaline for pancreatic enzymes and emulsifies fats

YEAR: 2023

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) Name the part of brain which is responsible for the following actions: (i) Maintaining posture and balance (ii) Beating of heart (iii) Thinking (iv) Blood pressure (b) Where are auxins synthesized in a plant? Which organ of the plant shows: (i) Positive phototropism (ii) Negative geotropism (iii) Positive hydrotropism

OPTIONS: (A) (i) Cerebellum (ii) Medulla (iii) Cerebrum (iv) Medulla; Tip of shoot; (i) Shoot/stem (ii) Shoot/stem (iii) Roots (B) (i) Cerebrum (ii) Medulla (iii) Cerebellum (iv) Medulla; Tip of root; (i) Roots (ii) Roots (iii) Shoot (C) (i) Cerebellum (ii) Cerebrum (iii) Medulla (iv) Cerebellum; Tip of shoot; (i) Roots (ii) Shoot (iii) Roots (D) (i) Medulla (ii) Cerebellum (iii) Cerebrum (iv) Medulla; Tip of shoot; (i) Shoot (ii) Roots (iii) Roots

ANSWER: (A) (i) Cerebellum (ii) Medulla (iii) Cerebrum (iv) Medulla; Tip of shoot; (i) Shoot/stem (ii) Shoot/stem (iii) Roots

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CHAPTER: 6 - Control and Coordination (Biology)

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YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which one of the given statements is incorrect?

OPTIONS: (A) DNA has the complete information for a particular characteristic (B) DNA is the molecule responsible for the inheritance of characters from parents to offsprings (C) Change in information will produce a different protein (D) Characteristics will remain the same even if protein changes

ANSWER: (D) Characteristics will remain the same even if protein changes

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Sensory nerve of a reflex arc carries information from the receptor cells to the:

OPTIONS: (A) spinal cord (B) brain (C) muscles of the effector organ (D) bones of the receptor organ

ANSWER: (A) spinal cord

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The hormone that regulates blood sugar level is secreted by:

OPTIONS: (A) Adrenal gland (B) Thyroid gland (C) Pancreas (D) Pituitary gland

ANSWER: (C) Pancreas

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The plant hormone that promotes cell division is:

OPTIONS: (A) Auxin (B) Cytokinin (C) Gibberellin (D) Absciscic acid

ANSWER: (B) Cytokinin

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: How is the brain and spinal cord protected in human beings? State one main function each of (i) Medulla and (ii) Cerebellum.

OPTIONS: (A) Brain is inside a bony box/skull/cranium; Vertebral column or backbone protects the spinal cord; Medulla: Controls involuntary actions such as blood pressure/salivation/vomiting; Cerebellum: Responsible for precision of voluntary actions/maintaining posture and balance (B) Brain is protected by the skull; Spinal cord is protected by the rib cage; Medulla: Controls voluntary actions; Cerebellum: Controls thinking (C) Brain is protected by the skull; Spinal cord is protected by the vertebral column; Medulla: Controls thinking; Cerebellum: Controls involuntary actions (D) Brain is protected by the cranium; Spinal cord is protected by the backbone; Medulla: Controls balance; Cerebellum: Controls blood pressure

ANSWER: (A) Brain is inside a bony box/skull/cranium; Vertebral column or backbone protects the spinal cord; Medulla: Controls involuntary actions such as blood pressure/salivation/vomiting; Cerebellum: Responsible for precision of voluntary actions/maintaining posture and balance

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A plant hormone helps in bending of stem towards light. Name the hormone and explain how is the movement achieved.

OPTIONS: (A) Auxin; When light is coming from one side, auxin diffuses towards the shady side of the shoot. This stimulates the cells of the shoot to grow longer and bend towards light (B) Gibberellin; It promotes cell elongation on the illuminated side causing

bending (C) Cytokinin; It promotes cell division on the illuminated side causing bending
(D) Absciscic acid; It inhibits growth on the illuminated side causing bending

ANSWER: (A) Auxin; When light is coming from one side, auxin diffuses towards the shady side of the shoot. This stimulates the cells of the shoot to grow longer and bend towards light

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: List two differences between the movement of leaves of a sensitive plant and the movement of a shoot towards light.

OPTIONS: (A) Sensitive plant: Non-directional, due to touch, fast, reversible; Shoot towards light: Directional, due to light, slow, irreversible (B) Sensitive plant: Directional, due to light, slow, irreversible; Shoot towards light: Non-directional, due to touch, fast, reversible (C) Sensitive plant: Due to growth, irreversible; Shoot towards light: Due to turgor changes, reversible (D) Sensitive plant: Slow, reversible; Shoot towards light: Fast, irreversible

ANSWER: (A) Sensitive plant: Non-directional, due to touch, fast, reversible; Shoot towards light: Directional, due to light, slow, irreversible

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Name the hormone released and the gland which secretes it in human beings during scary situations. How does the body respond to enable it to deal with the situation?

OPTIONS: (A) Adrenaline hormone; Adrenal gland; Heart beats faster, breathing rate increases, blood supply to digestive system and skin reduced, blood supply diverted to skeletal muscles (B) Thyroxine; Thyroid gland; Metabolism increases, heart beats faster (C) Insulin; Pancreas; Blood sugar level increases (D) Growth hormone; Pituitary; Body grows faster

ANSWER: (A) Adrenaline hormone; Adrenal gland; Heart beats faster, breathing rate increases, blood supply to digestive system and skin reduced, blood supply diverted to skeletal muscles

YEAR: 2023

CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Some plants like pea plants have tendrils which help them to climb up other plants. Explain how is it done. Name the plant hormone responsible for this movement.

OPTIONS: (A) Tendrils are sensitive to touch, when they come in contact with any support, the part in contact does not grow as rapidly as the part away from the object. This causes the tendril to circle around the object and cling to it; Auxin (B) Tendrils are sensitive to light, when they come in contact with any support, they grow towards it; Gibberellin (C) Tendrils are sensitive to water, when they come in contact with any support, they grow towards it; Cytokinin (D) Tendrils are sensitive to gravity, when they come in contact with any support, they grow away from it; Auxin

ANSWER: (A) Tendrils are sensitive to touch, when they come in contact with any support, the part in contact does not grow as rapidly as the part away from the object. This causes the tendril to circle around the object and cling to it; Auxin

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CHAPTER: 7 - Reproduction (Biology)

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YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Plants which bear unisexual flowers are:

OPTIONS: (A) Mustard and Papaya (B) Hibiscus and Watermelon (C) Mustard and Hibiscus (D) Watermelon and Papaya

ANSWER: (D) Watermelon and Papaya

YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Choose the correct order of the stages of binary fission in Leishmania.

OPTIONS: (A) I, II, III, IV, V (B) I, III, II, V, IV (C) I, III, V, II, IV (D) I, II, III, V, IV

ANSWER: (B) I, III, II, V, IV

YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Consider the following three flowers namely X, Y and Z. Which flower(s) would develop into a fruit?

OPTIONS: (A) X only (B) Z only (C) X and Y only (D) Y and Z

ANSWER: (C) X and Y only

YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The process of fusion of male and female gametes is called:

OPTIONS: (A) Pollination (B) Fertilisation (C) Germination (D) Fragmentation

ANSWER: (B) Fertilisation

YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Name the parts of a bisexual flower that are not directly involved in reproduction. (b) Differentiate between self pollination and cross pollination. List any two significance of pollination. (c) What is the fate of ovules and ovary after fertilization in a flower?

OPTIONS: (A) Sepals and petals; Self pollination: pollen from same flower, Cross pollination: pollen from different flower; Significance: leads to fertilization, produces seeds; Ovules → seeds, Ovary → fruit (B) Sepals and petals; Self pollination: pollen

from different flower, Cross pollination: pollen from same flower; Significance: produces seeds, helps in dispersal; Ovules → fruit, Ovary → seeds (C) Petals and stamens; Self pollination: pollen from same flower, Cross pollination: pollen from different flower; Significance: leads to fertilization; Ovules → seeds, Ovary → fruit (D) Sepals and petals; Self pollination: pollen from same flower, Cross pollination: pollen from different flower; Significance: leads to fertilization; Ovules → fruit, Ovary → seeds
ANSWER: (A) Sepals and petals; Self pollination: pollen from same flower, Cross pollination: pollen from different flower; Significance: leads to fertilization, produces seeds; Ovules → seeds, Ovary → fruit

YEAR: 2023

CHAPTER: 7 - Reproduction

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) What happens when the egg is not fertilised? (b) How is sperm genetically different from a human egg? (c) List any three contraceptive methods practised for family planning. Mention how these methods work.

OPTIONS: (A) The lining of uterus breaks down and comes out through vagina as blood and mucus; Sperm has X or Y chromosome, egg has only X chromosome; Surgical method - prevents sperm transfer; Barrier method - sperm does not reach egg; Oral pills - eggs are not released (B) The lining of uterus breaks down; Sperm has X chromosome only, egg has X or Y chromosome; Surgical method - prevents egg release; Barrier method - prevents sperm production; Oral pills - kills sperm (C) The lining of uterus thickens; Sperm has X or Y chromosome, egg has only X chromosome; Surgical method - prevents egg release; Barrier method - prevents sperm transfer; Oral pills - kills sperm (D) The lining of uterus breaks down and comes out through vagina; Sperm has X or Y chromosome, egg has only X chromosome; Surgical method - prevents sperm transfer; Barrier method - prevents egg release; Oral pills - prevents sperm production

ANSWER: (A) The lining of uterus breaks down and comes out through vagina as blood and mucus; Sperm has X or Y chromosome, egg has only X chromosome; Surgical method - prevents sperm transfer; Barrier method - sperm does not reach egg; Oral pills - eggs are not released

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CHAPTER: 8 - Heredity and Evolution (Biology)

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YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cross between pea plant with white flowers (vv) and pea plant with violet flowers (VV) resulted in F₂ progeny in which ratio of violet (VV) and white (vv) flowers will be:

OPTIONS: (A) 1:1 (B) 2:1 (C) 3:1 (D) 1:3

ANSWER: (C) 3:1

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of chromosomes in parents and offsprings of a particular species undergoing sexual reproduction remain constant due to:

OPTIONS: (A) doubling of chromosomes after zygote formation (B) halving of chromosomes after zygote formation (C) doubling of chromosomes before gamete formation (D) halving of chromosomes at the time of gamete formation

ANSWER: (D) halving of chromosomes at the time of gamete formation

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In humans, if gene (B) is responsible for black eyes and gene (b) is responsible for brown eyes, then the colour of eyes of the progeny having gene combination Bb, bb or BB will be black only.

OPTIONS: (A) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion (B) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion (C) Assertion is true but Reason is false (D) Assertion is false but Reason is true

ANSWER: (D) Assertion is false but Reason is true

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When an X bearing sperm fertilises the egg, the resulting zygote has the following combination of chromosomes:

OPTIONS: (A) 44 + XX (B) 44 + XY (C) 22 + XX (D) 22 + XY

ANSWER: (A) 44 + XX

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a dominant trait in pea plants?

OPTIONS: (A) Dwarf height (B) Wrinkled seeds (C) Yellow seeds (D) Green seeds

ANSWER: (C) Yellow seeds

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Mendel crossed a pure tall pea plant (TT) with a pure short pea plant (tt) and obtained all tall plants in F₁ generation. (a) What is the gene combination present in the plants of F₁ generation? (b) Give reason why only tall plants are observed in F₁ progeny. (c) What will be the ratio of the plants obtained in the F₂ generation when F₁ plants are self-pollinated?

OPTIONS: (A) Tt; Because only Dominant trait (Tall) is expressed in F₁ generation / Tallness is dominant over recessive short trait; F₂ generation – Tall : short = 3 : 1 (B) TT; Because only Recessive trait is expressed; F₂ generation – Tall : short = 1 : 1 (C) tt; Because only Dominant trait is expressed; F₂ generation – Tall : short = 1 : 3 (D) Tt; Because only Recessive trait is expressed; F₂ generation – Tall : short = 3 : 1

ANSWER: (A) Tt; Because only Dominant trait (Tall) is expressed in F₁ generation / Tallness is dominant over recessive short trait; F₂ generation – Tall : short = 3 : 1

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Write any two pairs of contrasting characteristics of pea plant used by Mendel other than those mentioned above. (b) Differentiate between dominant and recessive traits. (c) State the ratio of the combinations observed in the seeds of F₂ generation (in the above case). What do you interpret from this result?

OPTIONS: (A) Tall-Dwarf, White-Purple; Dominant Trait - expressed even if one copy exists; Recessive Trait - expression suppressed by dominant gene/expressed when two copies are present; 9:3:3:1; Traits are independently inherited (B) Tall-Dwarf, White-Purple; Dominant Trait - expressed only when two copies exist; Recessive Trait - expressed even if one copy exists; 3:1; Traits are linked (C) Round-Wrinkled, Yellow-Green; Dominant Trait - expressed even if one copy exists; Recessive Trait - expression suppressed by dominant gene; 9:3:3:1; Traits are independently inherited (D) Tall-Dwarf, White-Purple; Dominant Trait - expressed even if one copy exists; Recessive Trait - expressed when two copies are present; 3:1; Traits are independently inherited

ANSWER: (A) Tall-Dwarf, White-Purple; Dominant Trait - expressed even if one copy exists; Recessive Trait - expression suppressed by dominant gene/expressed when two copies are present; 9:3:3:1; Traits are independently inherited

YEAR: 2023

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Given below is a cross between a pure violet flowered pea plant (V) and a pure white flowered pea plant (v). Diagrammatically explain what type of progeny is obtained in F₁ generation and F₂ generation.

OPTIONS: (A) F₁: All violet (Vv); F₂: Violet:White = 3:1 (B) F₁: All white (vv); F₂: Violet:White = 1:3 (C) F₁: All violet (VV); F₂: Violet:White = 3:1 (D) F₁: All white (vv); F₂: Violet:White = 3:1

ANSWER: (A) F₁: All violet (Vv); F₂: Violet:White = 3:1

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CHAPTER: 9 - Light - Reflection and Refraction (Physics)

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YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: To obtain a magnification of +2 with a concave mirror of radius of curvature 60 cm the object distance must be:

OPTIONS: (A) -90 cm (B) -45 cm (C) -30 cm (D) -15 cm

ANSWER: (D) -15 cm

YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror?

OPTIONS: (A) Less than 10 cm (B) 10 cm (C) between 10 cm and 20 cm (D) more than 20 cm

ANSWER: (A) Less than 10 cm

YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In torch lights and head lights of vehicles, the bulb is placed:

OPTIONS: (A) between the pole and the focus of the reflector (B) very near to the focus of the reflector (C) between the focus and centre of curvature of the reflector (D) at the centre of curvature of the reflector

ANSWER: (B) very near to the focus of the reflector

YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The power of a lens is measured in:
OPTIONS: (A) Metre (B) Dioptre (C) Watt (D) Joule
ANSWER: (B) Dioptre

YEAR: 2023
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The phenomenon of splitting of white light into its constituent colours is called:
OPTIONS: (A) Reflection (B) Refraction (C) Dispersion (D) Scattering
ANSWER: (C) Dispersion

YEAR: 2023
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following: (i) Distance of the image from the mirror (ii) Height of the image
OPTIONS: (A) $v = -36$ cm; $h' = -6$ cm (B) $v = -36$ cm; $h' = 6$ cm (C) $v = 36$ cm; $h' = -6$ cm (D) $v = -24$ cm; $h' = -4$ cm
ANSWER: (A) $v = -36$ cm; $h' = -6$ cm

YEAR: 2023
CHAPTER: 9 - Light - Reflection and Refraction
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Define power of a lens. The focal length of a lens is -10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from

the optical centre of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case?

OPTIONS: (A) Power of lens: Ability of a lens to converge or diverge light rays/Reciprocal of focal length in metre; It is diverging/concave lens; $P = -10\text{ D}$; Sign of magnification = + or positive (B) Power of lens: Ability to converge light; It is converging/convex lens; $P = +10\text{ D}$; Sign of magnification = - or negative (C) Power of lens: Ability to diverge light; It is diverging/concave lens; $P = -10\text{ D}$; Sign of magnification = - or negative (D) Power of lens: Reciprocal of focal length in metre; It is converging/convex lens; $P = +10\text{ D}$; Sign of magnification = + or positive

ANSWER: (A) Power of lens: Ability of a lens to converge or diverge light rays/Reciprocal of focal length in metre; It is diverging/concave lens; $P = -10\text{ D}$; Sign of magnification = + or positive

YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An object 4 cm in size is placed at 25 cm in front of a concave mirror of focal length 15 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Find the nature and size of the image.

OPTIONS: (A) $v = -37.5\text{ cm}$; Real and inverted; $h' = -6\text{ cm}$ (B) $v = 37.5\text{ cm}$; Virtual and erect; $h' = 6\text{ cm}$ (C) $v = -37.5\text{ cm}$; Real and erect; $h' = -6\text{ cm}$ (D) $v = 37.5\text{ cm}$; Virtual and inverted; $h' = 6\text{ cm}$

ANSWER: (A) $v = -37.5\text{ cm}$; Real and inverted; $h' = -6\text{ cm}$

YEAR: 2023

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) An object of height 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find: (i) focal length of the mirror, and (ii) magnification of the image formed with sign as per Cartesian sign convention.

OPTIONS: (A) $f = -50\text{ cm}$; $m = -1$ (B) $f = -100\text{ cm}$; $m = -1$ (C) $f = -50\text{ cm}$; $m = +1$ (D) $f = -100\text{ cm}$; $m = +1$

ANSWER: (A) $f = -50\text{ cm}$; $m = -1$

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CHAPTER: 10 - The Human Eye and the Colourful World (Physics)

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YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In human eye the part which allows light to enter into the eye is:

OPTIONS: (A) Retina (B) Pupil (C) Eye lens (D) Cornea

ANSWER: (D) Cornea

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The change in the focal length of an eye lens in human beings is caused by the action of:

OPTIONS: (A) optic nerves (B) ciliary muscles (C) retina (D) cornea

ANSWER: (B) ciliary muscles

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The colour of the sky appears blue due to:

OPTIONS: (A) Reflection of light (B) Refraction of light (C) Scattering of light (D)

Dispersion of light

ANSWER: (C) Scattering of light

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age. (i) Give reason for the above statement. (ii) Name the defect that is likely to arise in the eyes in such a condition. (iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

OPTIONS: (A) It is due to gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens; Presbyopia; Bifocal/Concave + Convex lens (B) It is due to excessive curvature of the eye lens; Myopia; Concave lens (C) It is due to elongation of the eyeball; Myopia; Convex lens (D) It is due to weakening of the retina; Presbyopia; Convex lens

ANSWER: (A) It is due to gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens; Presbyopia; Bifocal/Concave + Convex lens

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The colour of clear sky from the earth appears blue but from the space it appears black. Why?

OPTIONS: (A) Size of particles in the atmosphere is smaller than the wavelength of visible light, so they scatter light of shorter wavelengths i.e. blue. In space, there is no scattering of light due to absence of particles (B) The atmosphere absorbs all other colours and reflects blue. In space, there is no atmosphere (C) The blue colour is refracted more by the atmosphere. In space, there is no refraction (D) The atmosphere contains blue-coloured particles. In space, there are no particles

ANSWER: (A) Size of particles in the atmosphere is smaller than the wavelength of visible light, so they scatter light of shorter wavelengths i.e. blue. In space, there is no scattering of light due to absence of particles

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Observe the following diagram and answer the questions following it: (i) Identify the defect of vision shown. (ii) List its two causes. (iii) Name the type of lens used for the correction of this defect.

OPTIONS: (A) Myopia/Short Sightedness; Excessive curvature of eye lens, Elongation of eyeball; Concave lens/Diverging Lens (B) Hypermetropia/Far Sightedness; Focal length is too long, Size of eyeball is small; Convex lens/Converging lens (C) Myopia; Focal length is too long, Size of eyeball is small; Convex lens (D) Hypermetropia; Excessive curvature of eye lens, Elongation of eyeball; Concave lens

ANSWER: (A) Myopia/Short Sightedness; Excessive curvature of eye lens, Elongation of eyeball; Concave lens/Diverging Lens

YEAR: 2023

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A person cannot see distinctly the object placed beyond 5 m from his eyes. Name the defect of vision the person is suffering from. Draw a ray diagram to illustrate this defect. List its two possible causes. Name the lens used for the correction of this defect.

OPTIONS: (A) Myopia; Excessive curvature of eye lens, Elongation of eyeball; Concave lens/Diverging lens (B) Hypermetropia; Focal length is too long, Size of eyeball is small; Convex lens/Converging lens (C) Myopia; Focal length is too long, Size of eyeball is small; Convex lens (D) Hypermetropia; Excessive curvature of eye lens, Elongation of eyeball; Concave lens

ANSWER: (A) Myopia; Excessive curvature of eye lens, Elongation of eyeball; Concave lens/Diverging lens

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CHAPTER: 11 - Electricity (Physics)

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YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An electric kettle consumes 1 kW of electric power when operated at 220 V. The minimum rating of the fuse wire to be used for it is:

OPTIONS: (A) 1A (B) 2A (C) 4A (D) 5A

ANSWER: (D) 5A

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are:

OPTIONS: (A) (i) $I = Q/t$, (ii) $W = V/Q$ (B) (i) $Q = I \times t$, (ii) $W = V \times Q$ (C) (i) $Q = I/t$, (ii) $V = W/Q$ (D) (i) $I = Q/t$, (ii) $Q = V/W$

ANSWER: (B) (i) $Q = I \times t$, (ii) $W = V \times Q$

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the:

OPTIONS: (A) ammeter is connected in parallel and the voltmeter in series (B) ammeter is connected in series and the voltmeter in parallel (C) ammeter and voltmeter are both connected in series (D) ammeter and voltmeter are both connected in parallel

ANSWER: (B) ammeter is connected in series and the voltmeter in parallel

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be:

OPTIONS: (A) 100% (B) 200% (C) 300% (D) 400%

ANSWER: (C) 300%

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a resistive circuit if the current is increased to two times, the percentage change in the amount of heat dissipated in the circuit would be:

OPTIONS: (A) 400% (B) 300% (C) 200% (D) 100%

ANSWER: (B) 300%

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be:

OPTIONS: (A) 0.04A (B) 0.06A (C) 0.08A (D) 0.12A

ANSWER: (B) 0.06A

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio R_s/R_p is:

OPTIONS: (A) 32 (B) 2 (C) 0.5 (D) 16

ANSWER: (D) 16

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it? State the law which helps us in solving the above stated question.

OPTIONS: (A) Current becomes one fourth of its original value; According to Ohm's law – potential difference is directly proportional to the current flowing through the conductor provided temperature remains constant (B) Current becomes one fourth of its original value; According to Ohm's law – potential difference is inversely proportional to the current (C) Current becomes four times; According to Ohm's law – potential difference is directly proportional to the current (D) Current becomes half; According to Ohm's law – potential difference is directly proportional to the current

ANSWER: (A) Current becomes one fourth of its original value; According to Ohm's law – potential difference is directly proportional to the current flowing through the conductor provided temperature remains constant

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Three resistors of 6 ohm, 4 ohm and 4 ohm are connected together so that the total resistance is 8 ohm. Draw a diagram to show this arrangement and give reason to justify your answer.

OPTIONS: (A) 4 ohm and 4 ohm in parallel (2 ohm) connected in series with 6 ohm (total 8 ohm) (B) 6 ohm and 4 ohm in parallel connected in series with 4 ohm (C) All three in series (14 ohm) (D) All three in parallel (1.5 ohm)

ANSWER: (A) 4 ohm and 4 ohm in parallel (2 ohm) connected in series with 6 ohm (total 8 ohm)

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: An electric iron of 1500 W, 200 V and a flash light of 500 W, 200 V are used in homes. The rating of fuse to be used should be:

OPTIONS: (A) 5A (B) 10A (C) 15A (D) 20A

ANSWER: (B) 10A

YEAR: 2023

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and 330 W when the heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case. (b) What is heating effect of electric current? (c) Find an expression for the amount of heat produced when a current passes through a resistor for some time.

OPTIONS: (A) Max: $I = 4A$, $R = 55 \text{ ohm}$; Min: $I = 1.5A$, $R = 146.6 \text{ ohm}$; When electric current is passed through a resistor, electrical energy is dissipated and appears as heat energy; $H = I^2Rt$ / $H = VIt$ (B) Max: $I = 2A$, $R = 110 \text{ ohm}$; Min: $I = 1.5A$, $R = 146.6 \text{ ohm}$; When electric current is passed through a resistor, electrical energy is dissipated; $H = VIt$ (C) Max: $I = 4A$, $R = 55 \text{ ohm}$; Min: $I = 3A$, $R = 73.3 \text{ ohm}$; When electric current is passed through a resistor, heat is produced; $H = I^2Rt$ (D) Max: $I = 4A$, $R = 55 \text{ ohm}$; Min: $I = 1.5A$, $R = 146.6 \text{ ohm}$; When electric current is passed through a resistor, electrical energy is dissipated and appears as heat energy; $H = IRt$

ANSWER: (A) Max: $I = 4A$, $R = 55 \text{ ohm}$; Min: $I = 1.5A$, $R = 146.6 \text{ ohm}$; When electric current is passed through a resistor, electrical energy is dissipated and appears as heat energy; $H = I^2Rt$ / $H = VIt$

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CHAPTER: 12 - Magnetic Effects of Electric Current (Physics)

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YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a current in a long straight solenoid, N and S poles are created at the two ends. Among the following statements, the incorrect statement is:

OPTIONS: (A) The magnetic field lines inside the solenoid are in the form of straight lines, which indicates that the magnetic field is uniform at all points inside the solenoid

(B) The strong magnetic field produced inside the solenoid can magnetize the soft iron placed inside it (C) The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet (D) The N and S poles exchange positions when the direction of current through the solenoid is reversed

ANSWER: (C) The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet

YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A constant current flows in a horizontal wire in the plane of the paper from east to west as shown in the figure. The direction of the magnetic field will be north to south at a point:

OPTIONS: (A) directly above the wire (B) directly below the wire (C) located in the plane of the paper on the north side of the wire (D) located in the plane of the paper on the south side of the wire

ANSWER: (A) directly above the wire

YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A positron enters a uniform magnetic field at right angles to it as shown. The direction of force experienced by the positron will be:

OPTIONS: (A) to the right (B) to the left (C) into the page (D) out of the page

ANSWER: (C) into the page

YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The magnetic field inside a long straight current carrying solenoid:

OPTIONS: (A) is zero (B) decreases as we move towards its end (C) increases as we move towards its end (D) is same at all points

ANSWER: (D) is same at all points

YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: State the rule to determine the direction of a (a) magnetic field produced around a straight conductor carrying current and (b) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

OPTIONS: (A) (a) Right-Hand Thumb Rule: Hold the current carrying conductor in right hand, such that thumb indicates direction of current, then the fingers will wrap around conductor in the direction of field lines; (b) Fleming's Left Hand Rule: Stretch the forefinger, middle finger, and thumb of the left hand mutually perpendicular. If forefinger indicates direction of magnetic field, middle finger indicates direction of current, then thumb points in direction of motion or force (B) (a) Left-Hand Thumb Rule; (b) Right-Hand Rule (C) (a) Fleming's Right Hand Rule; (b) Fleming's Left Hand Rule (D) (a) Right-Hand Thumb Rule; (b) Right-Hand Rule

ANSWER: (A) (a) Right-Hand Thumb Rule: Hold the current carrying conductor in right hand, such that thumb indicates direction of current, then the fingers will wrap around conductor in the direction of field lines; (b) Fleming's Left Hand Rule: Stretch the forefinger, middle finger, and thumb of the left hand mutually perpendicular. If forefinger indicates direction of magnetic field, middle finger indicates direction of current, then thumb points in direction of motion or force

YEAR: 2023

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) (i) State the rule used to find the force acting on a current carrying conductor placed in a magnetic field. (ii) Given below are three diagrams showing entry of an electron in a magnetic field. Identify the case in which the force will be (1) maximum and (2) minimum respectively. Give reason for your answer.

OPTIONS: (A) (i) Fleming's left-hand rule; (ii) (1) Force is maximum in Fig (i) because direction of motion of electron/current is at right angle to magnetic field; (2) Force is

minimum in Fig (iii) because electron is moving parallel to the direction of magnetic field
(B) (i) Fleming's right-hand rule; (ii) (1) Force is maximum in Fig (ii); (2) Force is minimum in Fig (iii)
(C) (i) Fleming's left-hand rule; (ii) (1) Force is maximum in Fig (iii); (2) Force is minimum in Fig (i)
(D) (i) Right-hand thumb rule; (ii) (1) Force is maximum in Fig (i); (2) Force is minimum in Fig (iii)
ANSWER: (A) (i) Fleming's left-hand rule; (ii) (1) Force is maximum in Fig (i) because direction of motion of electron/current is at right angle to magnetic field; (2) Force is minimum in Fig (iii) because electron is moving parallel to the direction of magnetic field

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CHAPTER: 13 - Our Environment (Biology)
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YEAR: 2023
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Study the given figure of a Food web and identify the primary consumer in the food web:
OPTIONS: (A) Mice and Bear (B) Rabbit and Cat (C) Rabbit and Fox (D) Mice and Rabbit
ANSWER: (D) Mice and Rabbit

YEAR: 2023
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is a non-biodegradable waste?
OPTIONS: (A) Vegetable peels (B) Paper (C) Plastic (D) Cotton cloth
ANSWER: (C) Plastic

YEAR: 2023
CHAPTER: 13 - Our Environment
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The gas that causes depletion of ozone layer is:

OPTIONS: (A) Carbon dioxide (B) Methane (C) Chlorofluorocarbons (CFCs) (D) Sulphur dioxide

ANSWER: (C) Chlorofluorocarbons (CFCs)

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a food chain, the maximum energy is available at:

OPTIONS: (A) Producers (B) Primary consumers (C) Secondary consumers (D) Tertiary consumers

ANSWER: (A) Producers

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The process of accumulation of non-biodegradable chemicals at each trophic level is called:

OPTIONS: (A) Bioaccumulation (B) Biomagnification (C) Biodegradation (D) Biosynthesis

ANSWER: (B) Biomagnification

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: List two reasons to show that the existence of decomposers is essential in an ecosystem.

OPTIONS: (A) They help in the breakdown of organic matter into simple inorganic raw materials; Help in natural replenishment of nutrients in soil; Help in keeping the environment clean (B) They produce oxygen; Help in photosynthesis (C) They provide

food to consumers; Help in energy transfer (D) They absorb carbon dioxide; Help in reducing pollution

ANSWER: (A) They help in the breakdown of organic matter into simple inorganic raw materials; Help in natural replenishment of nutrients in soil; Help in keeping the environment clean

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: We do not clean ponds or lakes but an aquarium needs to be cleaned regularly. Why?

OPTIONS: (A) An aquarium is a man-made ecosystem in which natural cleansing agents like decomposers are not present; Ponds and lakes are natural ecosystems which have natural cleansing agents (B) An aquarium has more waste than ponds; Ponds have larger volume of water (C) An aquarium contains more fish than ponds; Ponds have natural filters (D) An aquarium is smaller than ponds; Ponds have more aquatic plants

ANSWER: (A) An aquarium is a man-made ecosystem in which natural cleansing agents like decomposers are not present; Ponds and lakes are natural ecosystems which have natural cleansing agents

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: State two reasons why human-made materials like plastics persist for a long time in the environment.

OPTIONS: (A) Plastics are non-biodegradable; Decomposers like bacteria and fungi cannot breakdown plastics into simpler substances; They are inert substances (B) Plastics are biodegradable; They are easily broken down by microorganisms (C) Plastics are soluble in water; They dissolve and persist (D) Plastics are organic compounds; They cannot be decomposed

ANSWER: (A) Plastics are non-biodegradable; Decomposers like bacteria and fungi cannot breakdown plastics into simpler substances; They are inert substances

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) (i) Why does a kitchen garden called an artificial ecosystem while a forest is considered to be a natural ecosystem? (ii) While designing an artificial ecosystem at home, write any two things to be kept in mind to convert it into a self-sustaining system. Give reason to justify your answer.

OPTIONS: (A) Kitchen Garden → A man made ecosystem/non-sustainable; Forest → Ecosystem maintained by nature/self-sustainable; In a jar containing water we can provide oxygen through a pump and add a few aquatic plants and animals to make it a self-sustaining system; Justification: Oxygen is replenished continuously; Aquatic plants serve as food (B) Kitchen Garden → Natural ecosystem; Forest → Man-made ecosystem; Provide sunlight and water; Justification: Plants need sunlight and water (C) Kitchen Garden → A man made ecosystem; Forest → Natural ecosystem; Add fish and plants; Justification: They provide food (D) Kitchen Garden → An ecosystem maintained by nature; Forest → Man-made ecosystem; Provide oxygen through a pump; Justification: Oxygen is needed

ANSWER: (A) Kitchen Garden → A man made ecosystem/non-sustainable; Forest → Ecosystem maintained by nature/self-sustainable; In a jar containing water we can provide oxygen through a pump and add a few aquatic plants and animals to make it a self-sustaining system; Justification: Oxygen is replenished continuously; Aquatic plants serve as food

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Write the percentage of (i) solar energy captured by the autotrophs and (ii) energy transferred from autotrophs to the next level in a food chain. (b) What are trophic levels? Why do different food chains in an ecosystem not have more than four to five trophic levels? Give reason.

OPTIONS: (A) (i) 1%; (ii) 10% energy; Every step of a food chain where transfer of energy occurs; Since the amount of available energy keeps on becoming less as we move to higher trophic level, very little usable energy remains after four trophic levels (B) (i) 10%; (ii) 1% energy; Every step of a food chain where transfer of energy occurs;

Due to increase in energy at higher trophic levels (C) (i) 1%; (ii) 10% energy; Every step of a food chain; Due to decrease in energy at higher trophic levels (D) (i) 2%; (ii) 10% energy; Every step of a food chain; Due to loss of energy at each level

ANSWER: (A) (i) 1%; (ii) 10% energy; Every step of a food chain where transfer of energy occurs; Since the amount of available energy keeps on becoming less as we move to higher trophic level, very little usable energy remains after four trophic levels

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If a harmful chemical enters in a food chain comprising peacock, plants, rats and snakes, which of these organisms is likely to have the highest concentration of the chemical in its body. Justify your answer. Name the process involved and define it.

OPTIONS: (A) Peacock; Because peacock is occupying the top level of this food chain and the concentration of these non-biodegradable chemicals increase as we move up in a food chain; Process involved - Biomagnification; The increase in the concentration of harmful chemicals which accumulate progressively in the successive trophic levels of a food chain (B) Plants; Because plants absorb the chemicals from soil; Bioaccumulation (C) Rats; Because rats eat plants; Biomagnification (D) Snakes; Because snakes eat rats; Bioaccumulation

ANSWER: (A) Peacock; Because peacock is occupying the top level of this food chain and the concentration of these non-biodegradable chemicals increase as we move up in a food chain; Process involved - Biomagnification; The increase in the concentration of harmful chemicals which accumulate progressively in the successive trophic levels of a food chain

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Explain how some harmful chemicals enter our bodies through the food chain. Why is the concentration of these harmful chemicals found to be maximum in human beings?

OPTIONS: (A) Some harmful substances like pesticides are used to protect crops. When these chemicals are washed down in the soil or water bodies, they are absorbed

by plants along with water and minerals and by animals from water. When we consume these food items, the pesticides enter our body. As human beings occupy the top level in any food chain, maximum concentration of these chemicals get accumulated in them (B) Harmful chemicals are present in the air and are inhaled by humans directly; They get concentrated in human lungs (C) Harmful chemicals are present in the water we drink; They get accumulated in our body (D) Harmful chemicals are produced in our body during metabolism; They get concentrated in our body

ANSWER: (A) Some harmful substances like pesticides are used to protect crops.

When these chemicals are washed down in the soil or water bodies, they are absorbed by plants along with water and minerals and by animals from water. When we consume these food items, the pesticides enter our body. As human beings occupy the top level in any food chain, maximum concentration of these chemicals get accumulated in them

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Why is damage to the ozone layer a cause for concern? (b) What steps are being taken to limit this damage?

OPTIONS: (A) UV rays reach the earth and cause ill effects like skin cancer in human beings; Minimize the use of CFCs; Forging on agreement to freeze CFC production at 1986 levels (B) UV rays reach the earth and cause global warming; Minimize the use of CFCs (C) UV rays reach the earth and cause skin cancer; Ban the use of all chemicals (D) UV rays reach the earth and cause skin cancer; Increase the use of CFCs

ANSWER: (A) UV rays reach the earth and cause ill effects like skin cancer in human beings; Minimize the use of CFCs; Forging on agreement to freeze CFC production at 1986 levels

YEAR: 2023

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Write one difference between biodegradable and non-biodegradable wastes. List two impacts of each type of the accumulated waste on environment if not disposed off properly.

OPTIONS: (A) Biodegradable wastes can be broken down by biological processes; Non-biodegradable wastes cannot be broken down by biological processes; Impact of biodegradable: Foul smell, Breeding place for carriers of diseases; Impact of non-biodegradable: Biological Magnification, Affect soil fertility (B) Biodegradable wastes cannot be broken down; Non-biodegradable wastes can be broken down; Impact of biodegradable: Soil pollution, Water pollution; Impact of non-biodegradable: Air pollution, Land pollution (C) Biodegradable wastes are produced by living organisms; Non-biodegradable wastes are produced by non-living things; Impact of biodegradable: Soil fertility, Water pollution; Impact of non-biodegradable: Soil pollution, Air pollution (D) Biodegradable wastes are organic; Non-biodegradable wastes are inorganic; Impact of biodegradable: Foul smell, Breeding place; Impact of non-biodegradable: Biological Magnification, Soil fertility

ANSWER: (A) Biodegradable wastes can be broken down by biological processes; Non-biodegradable wastes cannot be broken down by biological processes; Impact of biodegradable: Foul smell, Breeding place for carriers of diseases; Impact of non-biodegradable: Biological Magnification, Affect soil fertility